

Literature Review Draft: Continuum-Tipped Minimally Invasive Surgical Robotics

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Abstract

This draft literature review is structured for a thesis on a UR5e-based macro-micro robotic system with a flexible continuum tip for minimally invasive surgery (MIS). It consolidates relevant findings from prior work in the *Lachlan Report*, identifies which foundations remain valid, and highlights where the state of the art now requires updated evidence. The review prioritizes four technical threads central to this thesis: (1) tendon-driven continuum manipulator design and motion algorithms, (2) integration with a macro robot (UR5e), (3) collision detection and avoidance for safe operation in constrained anatomy, and (4) rapid end-effector exchange mechanisms. It also includes explicit “missing area” headings to direct next-stage searching, reading, and synthesis.

1 Project Context and Problem Significance

This thesis extends prior macro-micro robotic surgery work by coupling a UR5e macro manipulator with a new flexible continuum tip, then improving movement algorithms inherited from earlier implementations. The applied significance is strong: MIS demands dexterity in constrained spaces, low tissue interaction force, high positional repeatability, and robust safety controls.

From the earlier project, the macro-micro concept and teleoperation feasibility are already established at a proof-of-concept level. The key opportunity now is to move from demonstration toward a more clinically credible engineering argument through stronger modeling, safety layers (especially collision handling), and modular hardware architecture (rapid tool exchange).

2 Inherited Knowledge from Lachlan Report: What Is Still Useful

The prior report remains highly relevant in these areas:

2.1 Continuum and Snake-like Design Foundations

The prior review appropriately positioned tendon-driven continuum/snake-like manipulators as a practical and dominant pathway for surgical dexterity. The classification of continuum architectures and the design trade-off framing (dexterity, stiffness, manufacturability, miniaturization) remain useful to retain.

2.2 Macro-Micro Integration Rationale

The macro-micro argument is still strong: a rigid macro arm contributes workspace and gross positioning, while a flexible micro tip contributes distal dexterity and safer interaction in confined regions. This directly aligns with using the UR5e as the macro stage and a continuum tip as the micro stage.

2.3 Teleoperation and Controller Pipeline Lessons

The earlier implementation experience (ROS integration, latency considerations, and vibration observations) provides practical system-level lessons for architecture and control-loop design. Even if this thesis shifts focus from pure teleoperation toward autonomy-assisted control, these integration lessons are still relevant.

2.4 Kinematic Modeling as a Bottleneck

The prior work highlighted that continuum-tip performance quality is tightly linked to model quality and calibration quality. That finding should be retained as a central thesis narrative point: movement algorithm improvement must be justified through both model structure and validation methodology.

3 What Must Be Brought Up to Date

The earlier literature base appears strongest through roughly 2022 and should now be updated in the following directions.

3.1 Post-2022 Continuum Manipulator Control

Update with recent results on:

- Data-driven and learning-augmented continuum control (model correction, residual learning).
- Real-time model-based control under friction, hysteresis, and tendon coupling.
- Multi-objective control formulations balancing tracking error, smoothness, and safety constraints.

3.2 Safety-Critical Surgical Robot Control

The inherited report discusses precision and teleoperation, but safety layers should now be treated as first-class control objectives:

- Constraint-aware motion planning and control barrier function style methods.
- Contact-aware control and force-limited behavior near anatomical boundaries.
- Fault handling and safe-stop behavior for software and communication faults.

3.3 Collision Detection and Avoidance (Core Gap for This Thesis)

This is one of the most important expansion areas for your project. The review needs a dedicated synthesis of:

- Geometric collision detection for continuum bodies (not just rigid-link approximations).
- Environment representations (point cloud, mesh, signed distance field) suitable for intraoperative updates.
- Real-time avoidance strategies compatible with UR5e + continuum-tip control bandwidth.

3.4 Rapid End-Effector Change and Modular Interfaces

The prior conclusion already identifies end-effector replacement as cumbersome. This thesis should elevate this into an explicit engineering research stream:

- Mechanical quick-change couplers for sterile or semi-sterile workflows.
- Repeatable kinematic/electrical interface registration after each swap.
- Calibration and safety checks needed after end-effector exchange.

4 Focused Literature Synthesis for Current Thesis Objectives

4.1 Thread A: Continuum Tip Design for MIS on UR5e

The review should compare tendon-driven continuum options based on criteria directly linked to your implementation:

- Outer diameter and achievable curvature.
- Tendon routing complexity and friction sensitivity.
- Manufacturability (3D printing route, assembly complexity, tolerance sensitivity).
- Instrument channel requirements and compatibility with intended tasks.

4.2 Thread B: Movement Algorithm Improvement over Lachlan Baseline

Define baseline algorithm(s) from inherited work, then review candidate improvements under consistent metrics:

- Tracking performance (RMS error, peak error, settling behavior).
- Robustness to tendon slack, backlash, and parameter drift.
- Computational feasibility for real-time execution.

The literature section should explicitly map each candidate algorithm class to likely implementation effort and expected gain.

4.3 Thread C: Collision Detection and Avoidance in Constrained Anatomy

Your review should separate the problem into three layers:

- **Perception/model layer:** what representation of anatomy and instruments is available in real time?
- **Detection layer:** how are self-collision, tool-environment collision, and forbidden regions detected?
- **Response layer:** how does the controller react (hard stop, velocity scaling, local re-plan, haptic/visual warning)?

4.4 Thread D: Rapid End-Effector Exchange

This thread should connect mechanism design with control and safety:

- Mechanical interchange design options and locking reliability.
- Coupling repeatability and resulting pose uncertainty.
- Required post-swap validation routine before re-entry into operation.

5 Explicit Missing Research Areas (Add These as Search Targets)

The following subsection titles are intentionally phrased as missing-area headings to guide your next reading pass.

5.1 Missing Area 1: Benchmark Datasets and Standardized Evaluation Protocols for Continuum Surgical Robots

Need: common benchmarks to compare controller improvements credibly across studies.

5.2 Missing Area 2: Real-Time Collision Avoidance Methods Validated on Macro-Micro Systems with Flexible Distal Tools

Need: methods tested on systems structurally similar to UR5e + continuum tip.

5.3 Missing Area 3: Registration Drift and Recalibration After Quick End-Effector Exchange

Need: evidence on how repeated tool swaps affect kinematic accuracy and safety margins.

5.4 Missing Area 4: Safety Cases for Human-in-the-Loop MIS Robots with Shared Control

Need: practical safety argument structures for semi-autonomous assistance during tele-operation.

5.5 Missing Area 5: Tendon Wear, Sterilization Effects, and Lifecycle Reliability in Reconfigurable Continuum Tools

Need: long-horizon reliability evidence to support repeated clinical-style usage.

5.6 Missing Area 6: Task-Specific Performance Metrics Beyond Tip Tracking Error

Need: clinically meaningful metrics (procedure time, tissue interaction force proxies, collision near-miss counts).

6 Rubric-Aligned Coverage Notes (MMAN4951/MMAN9451 Interim Report)

6.1 Criterion 1: Literature Review (High Weighting)

To target Distinction/High Distinction bands, this review must do more than summarize papers. It should:

- Show conceptual links between manipulator design, control, safety, and modularity.
- Explicitly identify and justify knowledge gaps your thesis addresses.
- Include recent work (not only foundational papers).
- Critically compare methods instead of listing them.

6.2 Criterion 2: Research Question and Plan

Your final report should ensure the plan is narratively tied to the review:

- Each project task should be justified by a gap identified in this section.
- Include feasible milestones and contingency options (especially for hardware delays and calibration issues).

6.3 Criterion 3: Project-Dependent Preparations

The literature should foreshadow practical preparations you can evidence:

- UR5e safety/process training,
- software stack readiness,
- preliminary prototype tests,
- risk register for collision and tool-swap failure modes.

6.4 Criterion 4: Document Presentation

For marks protection, maintain:

- consistent citation formatting,
- clear figure/table captions,
- polished grammar and structured argument flow.

7 Proposed Next Reading and Writing Sequence

1. Build a paper matrix with columns: method class, hardware platform, control rate, collision strategy, modularity strategy, metrics, limitations.
2. Fill collision-avoidance and quick-change sections first (highest novelty for your topic).
3. Revisit inherited movement algorithm section and position your planned improvement as a direct response to documented limitations.
4. Close the review with a concise “identified gap to thesis objective” mapping table.

8 Working Thesis Gap Statement (Draft)

Current literature and prior project work support feasibility of macro-micro MIS manipulation, but leave a practical translation gap at the intersection of **accurate continuum movement control**, **real-time collision-safe operation**, and **fast, repeatable end-effector reconfiguration**. This thesis addresses that gap on a UR5e-based platform by improving inherited motion algorithms and integrating safety and modularity mechanisms necessary for repeatable experimental deployment.